Choose a subject area and a specific problem. Use both structured data and unstructured text to build one knowledge graph (KG). Complete a proposal and two connected tasks.

Schedule and points

Stage	Discussion	Deadline	Points
Project proposal	October 28	November 3	20
Task 1	November 17	December 1	20
Task 2	December 15	Credit day in December	40
Total			80

Checkpoint 1 (November 3) covers the proposal: 20 points. Checkpoint 2 (the December credit day) covers Task 1 and Task 2: 60 points. Exam and individual-achievement points are not discussed here.

All-or-nothing grading

Each stage is assessed separately. To pass, complete all required work, prepare the required materials and explain your work at the defense. A passed stage earns its full score; a failed stage earns 0.

Passing a stage after its deadline halves its score: 10 points for the proposal, 10 for Task 1 and 20 for Task 2. An incomplete stage still earns 0.

Task 2 does not require perfect extraction or entity-matching accuracy. It requires the working methods, checks and analysis described below, there is no fixed accuracy threshold.

Data, scope and tools

Choose your subject area, sources, graph model and tools. Your schema must define entity types, relationships and properties. You may reuse an ontology or vocabulary or design your own schema. Explain your choices and cite reused material.

Your project must combine structured data and unstructured text in one knowledge graph. Use structured records for Task 1. In Task 2, extract additional entities and relationships from text; simply storing the text is not enough. Both types may be in one file, for example metadata columns and abstracts. If the dataset has only structured fields, add a relevant text source.

There is no fixed minimum number of files, documents or nodes. Agree the data volume, schema, evaluation sample and submission materials with the teacher. The data must support meaningful relationships, useful queries and checks against the sources.

A notebook or command-line program is sufficient. A ready-made graph, screenshots or a chatbot alone are not enough. Optional interfaces or advanced models do not replace the required work.

Project proposal and implementation plan — 20 points

Before implementation, present a plan for both tasks. Suggested length: 3–4 pages, including a draft schema diagram. Include sample records and text passages, and show how five source facts fit your schema.

1. Problem and scope. Identify the intended user, the problem and the project boundaries. Explain why relationships between entities matter.

2. Questions. Write five questions and state the expected answers and how you will check them. At least two must follow two or more relationships between domain entities; at least one must combine structured information with facts extracted from text. State which questions each task will answer.

3. Data. Describe the structured records and unstructured texts, their sources, examples, planned volume and access restrictions. Explain how you will connect them in the graph.

4. Schema. Show entity types, relationships and directions, properties, stable identifiers and links to sources. Explain how you will handle alternative or ambiguous names.

5. Work plan. Describe the steps, schedule, tools, intended LLM use and quality checks. Plan a human-checked evaluation sample and a fallback if a source or tool is unavailable.

For example, “Which courses teach the skills needed for a job role?” connects courses, skills and roles. Linking all entities to one source document does not count as a question that follows domain relationships.

Task 1. Build the initial knowledge graph — 20 points

Build the graph from structured or semistructured data. Provide code and instructions that another person can use to rebuild it.

1. Design. Define the schema and stable identifiers. Explain key choices, including when an item is a node rather than a property.

2. Build. Write scripts or a notebook that converts source records into nodes, relationships and properties. Handle missing values, duplicates, repeated imports and records that do not fit the schema.

3. Check. Record the source of each fact. Check identifier uniqueness, required properties and whether relationships connect existing nodes of the allowed types. Report graph statistics and manually verify a sample of facts against the sources.

4. Query. Answer at least three planned questions, including one that follows two or more domain relationships. Provide runnable queries, saved results and explanations of any limitations.

5. Present and defend. Provide the Task 1 materials in the shared submission checklist below and explain your design, checks and queries.

Reusing data or an ontology is allowed, but show your own modeling, conversion and checking work. For a mainly textual domain, the instructor may approve a small starting table prepared manually from cited texts; automate further extraction in Task 2.

Task 2. Add and check knowledge from text — 40 points

Extend the Task 1 graph with entities and relationships extracted from unstructured text in the same subject area. Add information not already represented in the graph, link it to existing entities and measure the quality of the results.

1. Select documents or free-text fields and explain any cleaning or splitting into passages. Record sources and identifiers.

2. Use an LLM, other natural language processing methods, rules or a combination to extract entities and relationships. Follow the schema. Check output format, allowed types and support in the text; record rejected or uncertain results.

3. A mention is a name or phrase in text; an entity is the object or concept it refers to. Standardize names and link mentions to entities using identifiers, context, an entity list or an ontology. Handle alternative spellings and identical names for different entities. Leave uncertain matches unresolved.

4. Add the extracted information to the existing graph and keep the supporting passage for each fact. Show examples of merged records, separate records and unresolved matches, where present; explain any absent cases. A separate, unconnected graph does not meet this requirement.

5. Follow the evaluation procedure below. Compare your starting method (the baseline) with one justified change to extraction, name standardization or matching.

6.Revisit all five questions. Include at least two questions that follow multiple domain relationships and one answer that combines the initial structured data with text-derived facts. Compare results before and after expansion; explain questions the data still cannot answer.

The same spelling alone does not prove that mentions refer to the same entity. Two entities appearing in the same text does not prove another relationship between them. Label inferred facts separately from facts stated in the source, and record how they were derived.

Evaluation

Select representative passages from documents or free-text fields. Agree the sample size with the teacher and explain your selection. Manually identify and check all entities, relationships and entity matches covered by the schema.

Reserve the sample for final evaluation: do not use it for prompt examples or rule tuning. Define correct matches in advance, including entity identity and relationship type, direction and connected entities. Test the baseline and changed method on the same sample with the same scoring rules.

Report precision, recall and F1 separately for entity and relationship extraction, with counts of correct, incorrect and missed items. Precision measures how much extracted information is correct, recall measures how much reference information was found, F1 combines both. For entity matching, report correct matches, the total checked and unresolved cases. State which cases each calculation includes.

Explain representative mistakes and their effects on the graph or answers. Save outputs before manual correction or report results before and after correction separately. Report the actual comparison, even if the change does not improve the results, and explain the limits of a small sample.

Submission and individual defense

Prepare a public repository on Github and submit each task to it. A notebook may contain several items. Keep the Task 1 graph so it can be compared with Task 2.

Materials	What to include
For both tasks	Source data or access instructions; schema and identifiers; code; a saved graph or export; graph checks; runnable queries, saved results and explanations; a README with software requirements and rebuild steps.
Also for Task 2	Text or text-field sources; extraction and matching outputs; prompts or rules; reference annotations; metrics; baseline comparison; error analysis; before-and-after query results.

At the defense, be ready to trace a fact to its source, explain an entity match, run or explain a query and discuss an error. You may use LLMs for planning, coding, extraction, matching, queries and report editing. Name the tools and models, explain their roles, and record available versions, run dates, settings and example prompts. Save the exact outputs used to build the graph and provide a way to rebuild it without new model calls. Describe how you checked or corrected their outputs. You remain responsible for understanding the work. An LLM response is not a factual source: support graph facts with the selected data. A person must check the evaluation reference annotations.